Review for Test 1

This is just a quick note for the homework. Please double-check the textbook before using it.

Chapter 2: Dimensions and Fundamentals

Example 2-1:

Example 2–1. A rocket projectile has the following characteristics:

Initial mass	200 kg
Mass after rocket operation	130 kg
Payload, nonpropulsive structure, etc.	110 kg
Rocket operating duration	3.0 sec
Average specific impulse of propellant	240 sec

Determine the vehicle's mass ratio, propellant mass fraction, propellant flow rate, thrust, thrust-to-weight ratio, acceleration of vehicle, effective exhaust velocity, total impulse, and the impulse-to-weight ratio.

(1) Mass ratio: (Eq. 2–8) $\mathbf{MR} = m_f/m_0$
(mf: the final mass of the vehicle after the rocket has ceased to operate, 130-110; m0: mass before the rocket operation, 200-110)

(2) Propellant mass fraction: $\zeta = (m_0 - m_f)/m_0$

(3) Propellant flow rate: propellant mass / operation duration.

(4) Thrust: $F = I_s \dot{w}$:

(5) Effective exhaust velocity $c = I_s g_0$

(6) Total impulse: $I_t = I_s w$

Example 2–2. The following measurements were made in a sea level test of a solid propellant rocket motor:

Burn duration	40 sec
Initial mass before test	1210 kg
Mass of rocket motor after test	215 kg
Average thrust	62,250 N
Chamber pressure	7.00 MPa
Nozzle exit pressure	0.070 MPa
Nozzle throat diameter	0.0855 m
Nozzle exit diameter	0.2703 m

Determine $\dot{m}$, v_2, c^*, c, and I_s at sea level, and c and I_s at 1000 and 25,000 m altitude. Assume an invariant thrust and mass flow rate and negligible short start and stop transients.

(1) Mass flow rate: Total propellant used / duration time = (initial mass-final mass) / time

$$F = \dot{m}v_2 + (p_2 - p_3)A_2 \qquad (2\text{–}14)$$

(2) Exit velocity v2:

$$c^* = p_1 A_t / \dot{m} \qquad (2\text{–}18)$$

(3) Characteristic velocity c*:

(4) Effective exhaust velocity:

$$c = v_2 + (p_2 - p_3)A_2/\dot{m} \qquad (2\text{–}16)$$

(5) Specific impulse: $I_s = F/\dot{m}g_0$

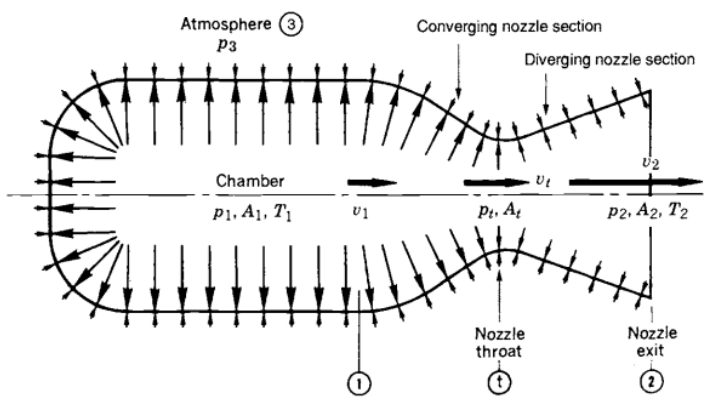

FIGURE 2–1. Pressure balance on chamber and nozzle interior walls is not uniform. The internal gas pressure (indicated by length of arrows) is highest in the chamber (p_1) and decreases steadily in the nozzle until it reaches the nozzle exit pressure p_2. The external or atmospheric pressure p_3 is uniform. At the throat the pressure is p_t. The four subscripts (shown inside circles) refer to the quantities A, v, T, and p at specific locations.

<u>Homework #3</u> (solved in metric in python & Excel)

Plot the variable of "Thrust" and "Specific Impulse" against "Altitude", using the atmospheric pressure information in the Appendix of your textbook. Use the "Minuteman" first-stage rocket engine information as follows:

- P_2 = 8.66 psia (assume to remain constant with altitude)
- Sea-Level Thrust, F = 194,600 lb
- Sea-Level Specific Impulse, I_{sp} = 254 seconds
- A_2 = 1,642 in^2

(1) and (2): Calculate effective exit velocity c and propellant mass flow m-dot:

$$c = I_s g_0 = F/\dot{m} \qquad (2\text{–}6)$$

(3) Calculate weight flow rate w-dot:

w-dot=m-dot * g0

(4) Calculate exit velocity v2 at sea level:

$$F = \dot{m}v_2 + (p_2 - p_3)A_2 \qquad (2\text{–}14)$$

Chapter 3: NOZZLE THEORY AND THERMODYNAMIC RELATIONS

HW#4:

(1) Reproduce Figure 3-1 when k=1.3. (The details is in Ch.3 Slides-1)

(2) Redo Example 3-1.

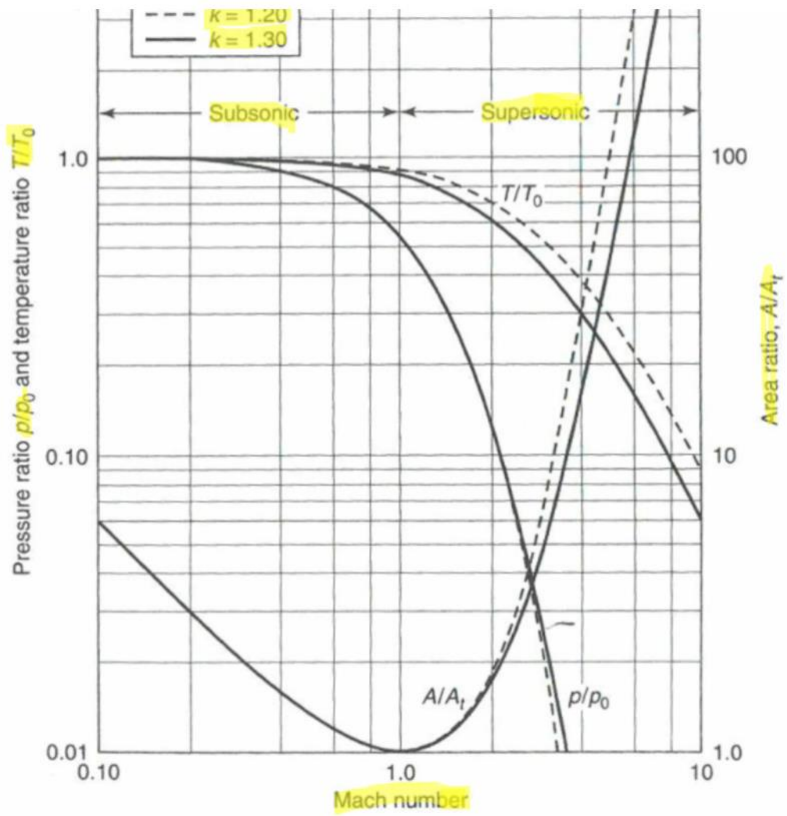

FIGURE 3–1. Relationship of area ratio, pressure ratio, and temperature ratio as functions of Mach number in a De Laval nozzle for the subsonic and supersonic nozzle

Isentropic relations:

$$\frac{A}{A^*} = \frac{1}{M} sqrt\left(\left(\frac{\left(1+\frac{g-1}{2}M^2\right)}{\frac{\gamma+1}{2}}\right)^{\frac{\gamma+1}{\gamma-1}}\right)$$

$$\frac{P}{P_0} = \left(1+\frac{\gamma-1}{2}M^2\right)^{-\frac{\gamma}{\gamma-1}}$$

$$\frac{T}{T_0} = \left(1+\frac{\gamma-1}{2}M^2\right)^{-1}$$

For optimum expansion:

- Nozzle exit pressure = ambient pressure
- Chamber pressure ≈ total pressure

$$p_0 = p\left[1+\tfrac{1}{2}(k-1)M^2\right]^{k/(k-1)}$$

$$= 0.1013\left[1+\tfrac{1}{2}\times 0.30\times 2.40^2\right]^{1.3/0.3} = 1.51\ \text{MPa}$$

Nozzle area ratio between the throat and exit can be calculated when M=1.

HW#5:

Example 3–2. A rocket operates at sea level ($p = 0.1013$ MPa) with a chamber pressure of $p_1 = 2.068$ MPa or 300 psia, a chamber temperature of $T_1 = 2222$ K, and a propellant consumption of $\dot{m} = 1$ kg/sec. (Let $k = 1.30$, $R = 345.7$ J/kg-K). Show graphically the variation of A, v, V, and M, with respect to pressure along the nozzle. Calculate the ideal thrust and the ideal specific impulse.

(1) Initial specific volume from the perfect gas law:

$$p_x V_x = RT_x \qquad (3\text{–}4)$$

(2) The specific volume at pressure of Px from isentropic flow process:

$$T_x/T_y = (p_x/p_y)^{(k-1)/k} = (V_y/V_x)^{k-1} \qquad (3\text{–}7)$$

$$V_x = V_1(p_1/p_x)^{1/k} =$$

$$T_x = T_1(p_x/p_1)^{(k-1)/k}$$

Or

(3) Exhaust velocity v2:

$$v_2 = \sqrt{\frac{2k}{k-1} RT_1 \left[1 - \left(\frac{p_2}{p_1}\right)^{(k-1)/k}\right]} = \sqrt{\frac{2k}{k-1} \frac{R'T_0}{\mathfrak{M}} \left[1 - \left(\frac{p_2}{p_1}\right)^{(k-1)/k}\right]} \qquad (3\text{–}16)$$

(4) Cross section area from the principle of mass conservation:

$$\dot{m}_x = \dot{m}_y \equiv \dot{m} = Av/V \qquad (3\text{–}3)$$

(5) Mach number:

$$M = v/a = v/\sqrt{kRT} \qquad (3\text{–}11)$$

(6) Thrust and impulse:

$$c = I_s g_0 = F/\dot{m} \qquad (2\text{–}6)$$

$$F = \dot{m}\, v_2$$

$$I_s = c/g_0$$

HW#6

Generate Fig. 3-4

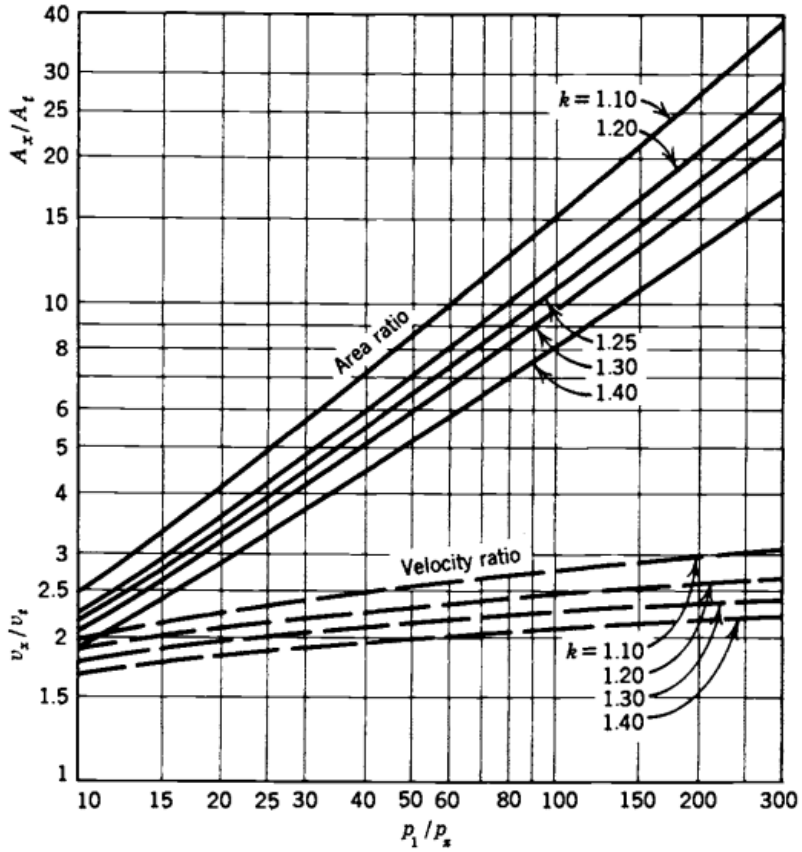

FIGURE 3–4. Area and velocity ratios as function of pressure ratio for the diverging section of a supersonic nozzle.

Area ratio (3-25) :

The ratio between the throat and any downstream area at which a pressure px prevails can be expressed as:

$$\frac{A_t}{A_x} = \frac{V_t v_x}{V_x v_t} = \left(\frac{k+1}{2}\right)^{1/(k-1)} \left(\frac{p_x}{p_1}\right)^{1/k} \sqrt{\frac{k+1}{k-1}\left[1 - \left(\frac{p_x}{p_1}\right)^{(k-1)/k}\right]} \qquad (3\text{–}25)$$

Velocity ratio (3-26)

$$\frac{v_x}{v_t} = \sqrt{\frac{k+1}{k-1}\left[1 - \left(\frac{p_x}{p_1}\right)^{(k-1)/k}\right]} \qquad (3\text{–}26)$$

HW#7

Example 3-3:

Example 3–3. Design a nozzle for an ideal rocket that has to operate at 25 km altitude and give 5000 N thrust at a chamber pressure of 2.068 MPa and a chamber temperature of 2800 K. Assuming that $k = 1.30$ and $R = 355.4$ J/kg-K, determine the throat area, exit area, throat velocity, and exit temperature.

(1) Critical (or throat) pressure:

$$p_t/p_1 = [2/(k+1)]^{k/(k-1)} \qquad (3\text{–}20)$$

(2) Critical (or throat) velocity:

$$v_t = \sqrt{\frac{2k}{k+1} RT_1} = a_t = \sqrt{kRT} \qquad (3\text{–}23)$$

(3) Ideal exit velocity:

$$v_2 = \sqrt{\frac{2k}{k-1} RT_1 \left[1 - \left(\frac{p_2}{p_1}\right)^{(k-1)/k}\right]} = \sqrt{\frac{2k}{k-1} \frac{R'T_0}{\mathfrak{M}} \left[1 - \left(\frac{p_2}{p_1}\right)^{(k-1)/k}\right]} \qquad (3\text{–}16)$$

(4) Ideal propellant consumption for optimum expansion condition: $\dot{m} = F/v_2$

(5) Specific volume at the entrance to the nozzle: $V_1 = RT_1/p_1$

(6) Specific volume at the throat:

$$V_t = V_1[(k+1)/2]^{1/(k-1)} \qquad (3\text{–}21)$$

(7) Specific volume at the exit for the isentropic flow process:

$$T_x/T_y = (p_x/p_y)^{(k-1)/k} = (V_y/V_x)^{k-1} \qquad (3\text{–}7)$$

$$V_2 = V_1 \left(\frac{p_1}{p_2}\right)^{1/k}$$

(8) The area at the throat and exit:

$$\dot{m}_x = \dot{m}_y \equiv \dot{m} = Av/V \qquad (3\text{–}3)$$

(where v is velocity and V is specific volume)

$$A_t = \dot{m}V_t/v_t$$

$$A_2 = \dot{m}V_2/v_2$$

(9) Nozzle area ratio (The ratio of the nozzle exit area A2 to the throat area At):

$$\epsilon = A_2/A_t \qquad (3\text{–}19)$$

(10) Exit temperature (for an isentropic flow process) :

$$T_x/T_y = (p_x/p_y)^{(k-1)/k} = (V_y/V_x)^{k-1} \qquad (3\text{–}7)$$

$$T_2 = T_1(p_2/p_1)^{(k-1)/k}$$